

Semester Project
Technical and Business Writing

Title: ML Based Cyber Threat Detection and Visualization Dashboard for Network Security

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Abstract

This proposal details the development of a Machine Learning (ML) system aimed at improving network security. Current security tools often fail to detect new types of attacks and overwhelm analysts with false alarms. This project proposes an ML framework to accurately classify network traffic as normal or malicious. The main gap we address is the lack of actionable intelligence: we will build an integrated dashboard to clearly visualize threats, making it easier and faster for security analysts to respond. The system will be evaluated using standard metrics like accuracy and False Positive Rate (FPR), aiming for high performance and practical usability in a real-world security environment.

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1. Introduction

1.1 Background and Problem Description

Modern organizations heavily depend on computer networks, making network security critical. Traditional security defenses, such as Signature-Based Intrusion Detection Systems (IDS), only identify threats they already know about. This makes them ineffective against new, or zero-day, attacks [1]. Furthermore, these systems generate too many alerts, leading to alert fatigue among security personnel. This high volume of unmanageable data presents a major problem in maintaining network integrity [2].

1.2 Proposed Solution Overview and Report Organization

We suggest an ML-based system to address this issue. Complex patterns of typical network behavior can be learned by machine learning models, which can then identify any deviations as possible threats [3]. Our solution consists of two primary components: an Interactive Visualization Dashboard for straightforward, real-time threat analysis and an ML Detection Engine for high-accuracy threat identification

The structure of this report is as follows: The specific research gap is identified in Section 2, the project's objectives are listed in Section 3, the background knowledge required is described in Section 4, the steps for system development are described in Section 5, the plan and expected results are summarized in Section 6, and ethical considerations and the conclusion are covered in Section 7.

1.3 Objective

The primary goal is to create an integrated ML-Based Cyber Threat Detection and Visualization Dashboard that improves operational effectiveness and network security performance.

Particular Goals:

1. On an unseen dataset, maintain a False Positive Rate (FPR) of less than 1% and attain a threat detection accuracy of more than 98%.
2. Create a reliable machine learning model (like Random Forest) to categorize multi-class network attacks.
3. Use a visualization dashboard that shows attack distribution, threat intensity, and temporal analysis in real time.

1.4 Scope of Work and Target Users

1. Functional Scope: Common network layer threats (such as DDoS, DoS, Brute Force, and Port Scan) will be the system's main focus. Instead of doing deep packet inspection, we will employ network flow data (metadata).

2. Target Users: Network administrators and Security Operations Center (SOC) analysts are the main users who want fast, accurate information to monitor network security.

2. Literature Review

2.1 ML Techniques for Network Security

The shift from traditional methods to ML has introduced two main approaches:

- Supervised Learning: Models like Random Forest (RF) are trained on labelled data (Attack vs. Normal). They are highly effective for known attack types [5].
- Unsupervised Learning: Models like Isolation Forest are trained only on 'normal' data. They are useful for detecting deviations, which is key for finding zero-day attacks[6].

2.2 Visualization Gap Analysis

Current network visualization tools often focus on network health (up/down status) or simply dump log data. They rarely integrate the real-time output of predictive ML models into a single, contextual view. This gap slows down human response, which is often the weakest link in security defense[4].

3. Problem Analysis & Research Gap

3.1 Limitations of Current Security Systems (IDS, Signature-Based)

The current limitation is two-fold:

1. Low Detection Rate for Novel Attacks: Signature-based IDS cannot recognize new attacks because they rely on fixed rule-sets [1].
2. Operational Inefficiency: High volumes of raw log data and False Positive Rates (FPR) mean security analysts waste time investigating harmless activity [2].

3.2 Identified Research Gap: Lack of Actionable Intelligence through Integrated Visualization

Research often stops after proving an ML model is accurate. The major gap in the industry is the lack of a system that seamlessly connects the complex ML output to the analyst's workflow [4]. Our project directly addresses this by building a dedicated dashboard that doesn't just list alerts but provides context (e.g., geographic source, severity score) visually, enabling analysts to make faster and better decisions.

4. Proposed Methodology

4.1 Dataset and Data Preprocessing Plan

- Dataset Selection: We will use the CICIDS 2017 dataset, as it contains modern, labeled attack types and comprehensive network flow features, providing a realistic training environment [7][8].

- Preprocessing: Managing missing values is part of data cleaning. In order to keep big values from controlling the model training process, features will be scaled using normalization. In order to utilize only the most essential network flow characteristics, we will also carry out feature selection[6].

4.2 Proposed ML Models

We will experiment with a blend of algorithms for comprehensive threat coverage:

1. Random Forest (RF): High-speed, high-accuracy supervised model for known attack classification [5].
2. Isolation Forest (IF): Unsupervised model to detect anomalies that represent novel threats [6].

4.3 System Architecture and Dashboard Approach

The system will follow a simple pipeline:

1. Data Ingestion: Network data is collected.
2. Preprocessing: Data is cleaned and features are extracted.
3. ML Engine: The trained models classify the traffic (Normal or Threat).
4. Database/Alerting: Classification results are stored and high-severity alerts are flagged.
5. Visualization Dashboard: This interface displays the data using tools like Plotly Dash or Streamlit. Key visualization features will include:
 - Real-time Threat Severity Score.
 - Geographic mapping of attack origins.
 - Time-series plots showing attack frequency (temporal analysis).

5. Expected Outcomes and Project Plan

5.1 Expected Outcomes

The successful completion of this project will result in:

1. A robust threat detection model meeting the high accuracy and low FPR objectives.
2. A functional, integrated dashboard that provides actionable, real-time insights to security analysts.
3. Demonstrated improvement in threat response time compared to manual log analysis.

5.2 Project Timeline and Deliverables

The project will be completed in four main phases as shown in Table I.

TABLE I

PROJECT TIMELINE

Phase	Description	Estimated Duration
Phase I	Data Preparation & Literature Review	4 Weeks
Phase II	ML Model Training & Optimization	6 Weeks
Phase III	Dashboard Development & System Integration	5 Weeks
Phase IV	Testing, Evaluation, and Final Documentation	3 Weeks

Deliverables: Trained ML Model, Functional Visualization Dashboard, Final Project Report.

6. Ethical Considerations and Conclusion

6.1 Ethical and Security Points

Only publicly accessible, encrypted datasets (CICIDS 2017) without personal data will be used. Data privacy is therefore guaranteed. The ML models will be created to reduce bias and guarantee only defensive, responsible use.

6.2 Conclusion

This proposal outlines a timely and necessary solution to tackle evolving cyber threats. By combining high-performance Machine Learning with user-centric visualization, this project aims to provide network security teams with the tools needed to move from reactive defense to proactive security intelligence.

7. References

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- [6] P. Agrawal, “End-to-End Network Anomaly Detection on CIC-IDS2017,” GitHub, 2024.

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[8] Canadian Institute for Cybersecurity, “IDS 2017 Dataset,” University of New Brunswick, 2025.

8. Glossary

Term	Simple Explanation
Actionable Intelligence	Useful information that tells a security person exactly what to do right now to fix a problem.
Alert Fatigue	Getting too many fake warnings (false alarms) which makes the security staff tired and causes them to miss the real danger.
CICIDS 2017	A popular set of internet traffic data used to train security programs to recognize attacks.
Deep Packet Inspection	Looking closely inside the content of an internet data piece (packet), not just the address, to check for bad stuff.
DDoS (Distributed Denial of Service)	A bad attempt to make a website or online service stop working by flooding it with too much data from many different computers.
DoS (Denial of Service)	A bad attempt to make a single computer or network stop working for normal users.
False Positive Rate (FPR)	The rate at which the security program makes a mistake by saying that normal activity is actually an attack.
Feature Selection	Picking only the most important pieces of information from the data to help the ML program learn better.
IDS (Intrusion Detection System)	A program or tool that watches the network to find and warn about suspicious activity.
Isolation Forest (IF)	A smart computer program that finds weird and unusual things happening in the network, which are often new attacks.
ML Engine	The main software part that holds the trained Machine Learning program and decides if network traffic is safe or dangerous.
Normalization	Making all the numbers in the data the same size range so the learning program treats them fairly.
Random Forest (RF)	A fast, reliable computer program that uses many simple decision rules to correctly identify known attacks.
Signature-Based Defense	An old security method that can only stop attacks it has seen before and has a specific rule (signature) for.
Temporal Analysis	Studying and showing how often attacks happen and when they happen using a timeline.

Zero-Day Attack	A broad term that describes recently discovered security vulnerabilities that hackers can use to attack systems.
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